

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7.	(a)	(i)	$v_{\text{plastic}} = \frac{AC}{BD} \times c$ or equivalent or by implication (1) BD = 44 or 45 or 46 [mm]; AC = 31 or 32 or 33 [mm] (1) $v_{\text{plastic}} = (2.1 \pm 0.1) \times 10^8 [\text{m s}^{-1}]$ (1)		3		3	3	
		(ii)	$n = 1.4 [\pm 0.1]$ ecf		1		1	1	
	(b)	(i)	A correct relevant time calculation e.g. for 120 m, 608 ns or for 120.9 m, 613 ns or by implication (1) Extra time by zig-zag route = 5 ns [4.6 ns] (1) Therefore overlap will occur, [as 4.6 ns > 4.0 ns] no ecf (1) [Allow 1 mark if n (refractive index) omitted leading to 3.0 ns delay and hence no overlap] Alternative: Extra distance for 4.0 ns extra time $\frac{3.00 \times 10^8 \times 4.0 \times 10^{-9}}{1.52}$ (1) = 0.79 [m] (1) Therefore overlap will occur [as 0.79 < 0.90 m] (1) [Allow 1 mark if n (refractive index) omitted leading to 1.2 m delay and hence no overlap]			3	3	2	
		(ii)	$n_{\text{clad}} [\sin 90^\circ] = 1.520 \sin 83^\circ$ or equivalent (1) $n_{\text{clad}} [\sin 90^\circ] = 1.51$ (1)		2		2	2	
			Question 7 total	0	6	3	9	8	0